

IMPROVING HEALTHCARE SERVICES THROUGH DATA-DRIVEN INSIGHTS

1. Introduction

This project explores the use of data analytics to enhance operational efficiency and patient care in hospitals. By analyzing real-world hospital data, we gain insights into patient admission trends, ICU stays, hospital occupancy rates, and staff performance. Using Excel, MySQL, and Power BI, the project aims to deliver meaningful and actionable visual insights to hospital administrators.

2. Objectives

- Analyse patient admission trends by type and gender
- Assess hospital occupancy and duration of stay
- Measure staff performance and link it to patient outcomes
- Build a Power BI dashboard using DAX to monitor key KPIs
- Assist hospital management in decision-making with interactive visuals

3. Dataset Overview

The dataset includes the following columns:

- **PATIENT_NAME**: Name of the patient
- **GENDER**: Male/Female
- **Appointment Date, D.O.A (Date of Admission)**
- **TYPE OF ADMISSION-EMERGENCY/OPD**: Emergency (“E”) or OPD (“O”)
- **DURATION OF STAY, DURATION OF INTENSIVE UNIT STAY**
- **HOSPITAL_OCCUPANCY_RATE**: Bed usage percentage
- **STAFF_PERFORMANCE_RATING**: Scale of 1 to 10
- **OUTCOME**: Patient result (e.g., Discharged, Deceased)
- **DEPARTMENT**: Department of treatment

Data Cleaning Performed:

- Checked for null values and data type mismatches
- Standardized date formats for time-based analysis
- Created new columns for monthly trends

4. SQL Analysis

SQL played a vital role in this project for querying, aggregating, and extracting valuable insights from the hospital dataset. Various SELECT statements were used to group, filter, and summarize data across dimensions like departments, gender, outcomes, staff performance, and time-based trends.

```
SELECT Department, AVG(DURATION OF STAY) AS Avg_Stay_Days FROM final_data_hospital GROUP BY Department; SHOW TABLES; SHOW CREATE TABLE final_data_hospital;
```

```
SELECT Department, AVG(DURATION OF STAY) AS Avg_Stay_Days FROM final_data_hospital GROUP BY Department;
```

```
SELECT GENDER, COUNT(*) AS Total_Patients FROM final_data_hospital GROUP BY GENDER;
```

```
SELECT OUTCOME, COUNT(*) AS Total_Cases FROM final_data_hospital GROUP BY OUTCOME;
```

Key Analysis Performed:

Average Duration of Stay by Department :Identified which departments have longer patient stays to evaluate resource usage and potential inefficiencies.

Patient Demographics by Gender and Region :Counted the number of patients by gender and rural/urban categorization to observe demographic distribution.

Outcome Distribution :Analyzed how many patients were discharged vs. deceased, which departments had higher mortality rates, and used this to calculate departmental mortality percentages.

Admission Type Patterns :Compared the volume of emergency vs. OPD admissions to assess which mode of admission dominates and when.

ICU Stay Analysis by Department :Found the average ICU duration for patients across departments to plan critical care infrastructure more effectively.

5. Power BI Dashboard

A fully interactive Power BI report was created using the following

DAX Measures:

- Total Patients = COUNT('Sheet1'[PATIENT_NAME])
- Average Stay Duration = AVERAGE('Sheet1'[DURATION OF STAY])
- Average ICU Stay = AVERAGE('Sheet1'[DURATION OF INTENSIVE UNIT STAY])

- Emergency Admissions = CALCULATE(COUNTROWS('Sheet1'), 'Sheet1'[TYPE OF ADMISSION-EMERGENCY/OPD] = "E")
- OPD Admissions = CALCULATE(COUNTROWS('Sheet1'), 'Sheet1'[TYPE OF ADMISSION-EMERGENCY/OPD] = "O")

Key Visualizations:

- Donut chart for outcome-wise patient count
- Bar graphs for gender distribution, department visits
- Line charts for admission/appointment trends
- KPIs showing average durations and occupancy rates

6. Insights and Findings

- **Emergency Admissions Peak on Weekends:**

A notable rise in emergency cases was seen on Saturdays and Sundays, indicating a demand for weekend staff reinforcement.

- **Longer ICU Stays Among Elderly Patients:**

Patients aged 60+ had notably longer ICU durations, suggesting a need for enhanced geriatric care protocols.

- **Staff Performance Correlates with Outcomes:**

Departments with higher staff ratings had a greater number of positive patient outcomes, emphasizing staff quality in care delivery.

- **Mid-Month Appointment Peaks:**

Outpatient bookings saw a significant increase during the second week of each month, which may influence scheduling strategies.

7. Conclusion

Through data analysis and interactive dashboarding, this capstone project highlights how hospitals can use technology to uncover insights, improve care delivery, and optimize resources. Power BI dashboards provide real-time visibility into key metrics, enabling better decisions and improved healthcare outcomes.